

Harrison Zuleta Montoya

+573052654536 | harrison.zmontoya@gmail.com | [linkedin.com/in/harrison-zuleta-montoya](https://www.linkedin.com/in/harrison-zuleta-montoya) | github.com/harrisonzm

Professional Summary

Systems Engineering student at the National University of Colombia, with more than 2 years of experience in backend development
Meta using Java (Spring Boot), TypeScript (NestJS, Express), and PostgreSQL/MongoDB databases. I have participated in projects with modular architectures, AWS deployment, and Scrum methodologies, prioritizing code quality, security, and scalability. Passionate about software engineering and building solutions that make an impact.

Skills

Programming Languages: Java, JavaScript, TypeScript, Python, Ruby, C, C++
Frameworks & Libraries: React, Angular, Svelte, Nextjs, Expressjs, Nestjs, Spring Boot, Fastapi, Django
Developer Tools: AWS, GCP, Docker

Experience

Software Developer <i>ABI (Connect AI)</i>	Oct 2025 — Nov 2025 <i>Remote</i>
• I achieved complete automation of quote delivery by orchestrating backend services, integrating WhatsApp and email APIs for dynamic delivery of generated PDFs. • I improved system modularity by injecting dependencies into TypeScript, refactoring backend logic with Tsyringe, and structuring repositories in TypeORM. • I optimized the user experience by creating interactive and responsive dashboards when developing components in Angular for broker, user, and role management. • I reduced delivery errors by 35% through real-time validations by implementing error control flows and forms with cross-validations. • I automated processes using backend scripts by integrating MessageBird and SMTP services.	
Generative AI Engineer <i>Freelance</i>	May 2025 — Aug 2025 <i>remote</i>
• I developed reflective multi-agent flows with LangGraph, reducing response time by 35% and increasing decision accuracy by 20% through routine testing, data analysis, and the development of optimized APIs for integration and user experience. • I achieved a fully functional multi-agent architecture by designing modular flows integrating LangGraph and LangChain for query classification, insight generation, and contextual data analysis. • I improved information retrieval accuracy by 30% by implementing a local RAG pipeline using ChromaDB and optimized prompting templates. • I strengthened system stability through iteration controls and advanced error handling by modularizing nodes and visualizing flows in LangGraph Studio. • I automated the export and synchronization of analytical results using a scalable output layer by connecting TXT/CSV artifacts to AWS S3 through boto3 and custom scripts.	
Fullstack Developer <i>Universidad Nacional de Colombia</i>	Aug 2024 — Aug 2025 <i>remote</i>
• I led the comprehensive design and implementation of the UNGameLab web platform, increasing the organization and reuse of educational resources by 50% and facilitating access for more than 150 teachers by centralizing, classifying, and sharing educational games. • I achieved a secure and scalable backend API by designing a modular architecture when implementing services with Express.js and NestJS, integrated with PostgreSQL and MongoDB. • I improved authentication and authorization mechanisms by implementing JWT and OAuth2 when designing a role-based access control system (user, creator, moderator).	

- I optimized data performance through asynchronous service orchestration by using Sequelize and Mongoose to manage complex relationships between entities.
- I strengthened the reliability and scalability of the system by automating continuous deployment through the implementation of APIs in AWS (EC2, S3) and Cloudflare, within a Scrum framework.

Head Software Engineer

Jan 2025 — Jan 2026

Agora Excelencia

- Lead the development, design and deployment of highly scalable solutions using modern development tools, optimizations and architecture to deliver reliable, valuable and appealing software platforms in an agile environment managing the requirement analysis, consolidation and establishments through assertive, supportive and comprehensive mindset with customers leading to all round solutions and design of working scalable and highly stable solutions.

Leadership Experience

Teaching Assistant – Operating Systems

Apr 2025 — Jan 2001

Universidad Nacional de Colombia

- I led teaching and academic support sessions, strengthening the understanding of more than 40 students in advanced operating system topics by planning sessions, providing guided exercise solutions, and giving individualized feedback.
- I consolidated practical learning in processes, threads, synchronization, memory, and file systems through programming exercises in C and Linux, using tools such as GDB, semaphores, mutexes, and virtual machines to simulate real environments.
- I supported the continuous evaluation of the course by grading and technically analyzing weekly assignments, applying rigorous performance criteria and providing feedback aimed at improving systems programming skills.

Projects

students and signatures administration system | *React, TypeScript, React Router DOM, MUI, Module Federation, NestJS, Docker*

- I developed a modular web platform that allows students to register, create courses, and enroll in real time, consuming data from independent microservices.
- I achieved real-time student enrollment and updates through an event-driven microservices system by implementing TCP communication between independent services in NestJS.
- I improved code scalability and maintainability by integrating microfrontends using Module Federation to reuse React components across projects.
- I strengthened the reliability and security of the API through exhaustive validations by applying class-validator and class-transformer within NestJS pipelines.
- I optimized the deployment and orchestration of services using independent containers by configuring Docker Compose to manage modular environments.
- I facilitated collaborative work between distributed teams through architectural decoupling by ensuring data and version independence between microservices.

Certifications

Java Backend with Spring

2025-10

Platzi

Credential ID: 784f90f1-ac8c-4097-9273-3b6f20c92d0e

Education

National university of Colombia

Systems and computer science engineering

Aug 2022 — Jan 2001